

Preliminary Data Analysis Report

Economic Uncertainty, Information Seeking, and Financial Markets: Evidence from Google Trends

1. Dataset Description

The final dataset contains 193 monthly observations from January 2010 to January 2026. It combines Germany Economic Policy Uncertainty (EPU), Google Trends Search Index, and DAX Volatility. Data were collected from PolicyUncertainty.com, Google Trends, and Yahoo Finance, then cleaned, transformed to monthly frequency, and merged into a unified dataset.

2. Descriptive Statistics

Table 1. Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
Germany EPU Index	338.49	283.00	65.22	1502.18
Search Index	22.73	10.42	13.60	90.80
DAX Volatility	0.0110	0.0054	0.0040	0.0471

Germany EPU exhibits substantial variation over the sample period, with values ranging from 65.22 to 1502.18. The Search Index also shows noticeable fluctuations, while DAX Volatility remains relatively stable except during periods of market stress.

3. Correlation Analysis

Table 2. Pearson Correlation Matrix

Variable	Germany EPU Index	Search Index	DAX Volatility
Germany EPU Index	1.000	0.626	0.027
Search Index	0.626	1.000	0.092
DAX Volatility	0.027	0.092	1.000

The strongest relationship is observed between Germany EPU and the Search Index ($r = 0.626$), suggesting that periods of higher economic policy uncertainty are associated with increased information-seeking behavior. In contrast, the correlations involving DAX Volatility are relatively weak, indicating that market volatility may be influenced by additional factors beyond policy uncertainty.

4. Visual Analysis

Figure 1. Germany Economic Policy Uncertainty (EPU), 2010–2026

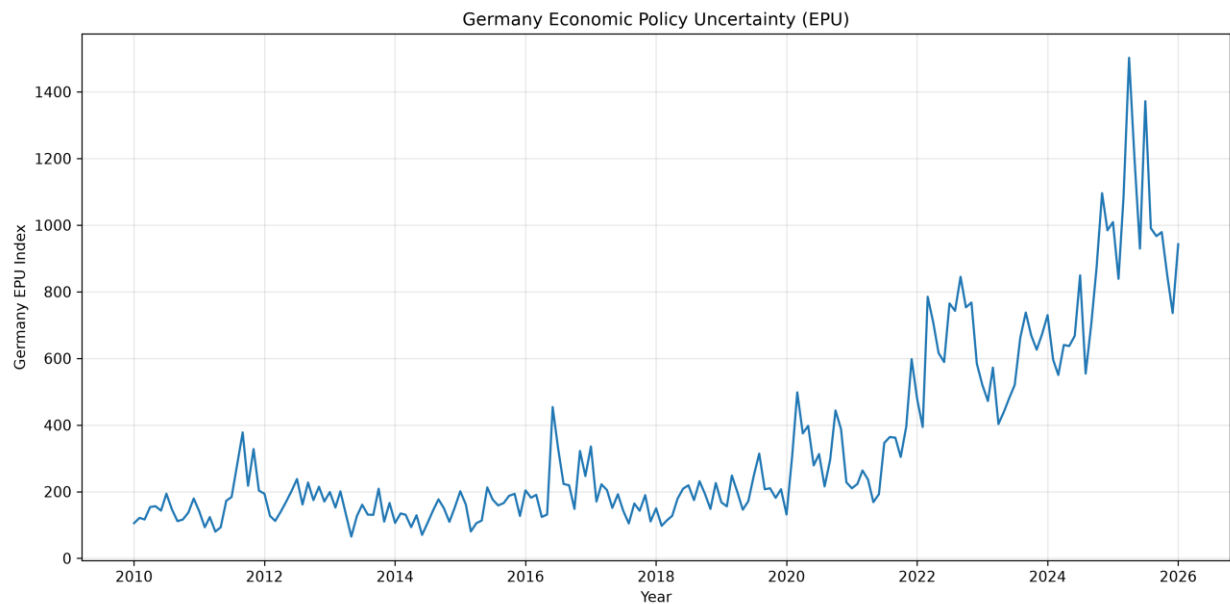
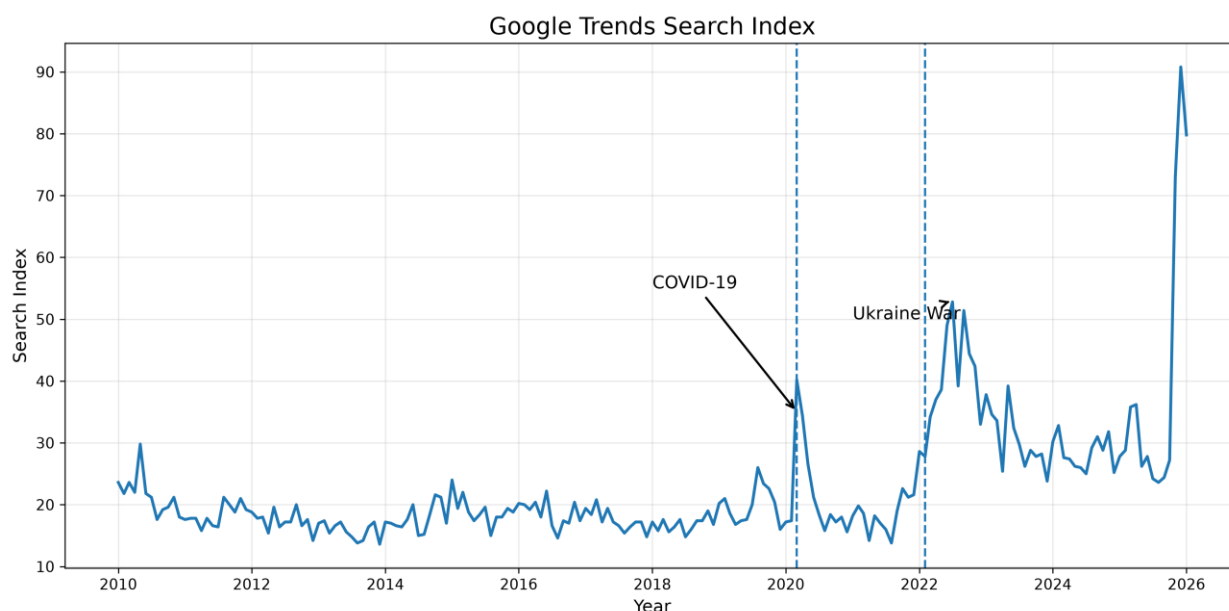


Figure 1 shows the evolution of Germany’s Economic Policy Uncertainty (EPU) index from 2010 to 2026. Several major spikes coincide with important economic and geopolitical events,

including the Eurozone debt crisis (2011–2012), the COVID-19 pandemic (2020), and the Ukraine war and energy crisis (2022). These events increased uncertainty regarding economic policy and future economic conditions in Germany.

Note: Vertical dashed lines indicate major economic and geopolitical events associated with increases in economic policy uncertainty.

Figure 2. Google Trends Search Index, 2010–2026



Search activity generally increases during periods of elevated uncertainty, particularly during the COVID-19 pandemic and the Ukraine war. Overall, the pattern provides preliminary evidence that uncertainty is associated with increased information-seeking behavior. Further analysis will examine whether these changes are systematically related to movements in economic policy uncertainty.

Note: Vertical dashed lines indicate major economic and geopolitical events associated with increased uncertainty.

Figure 3. Monthly DAX Volatility, 2010–2026

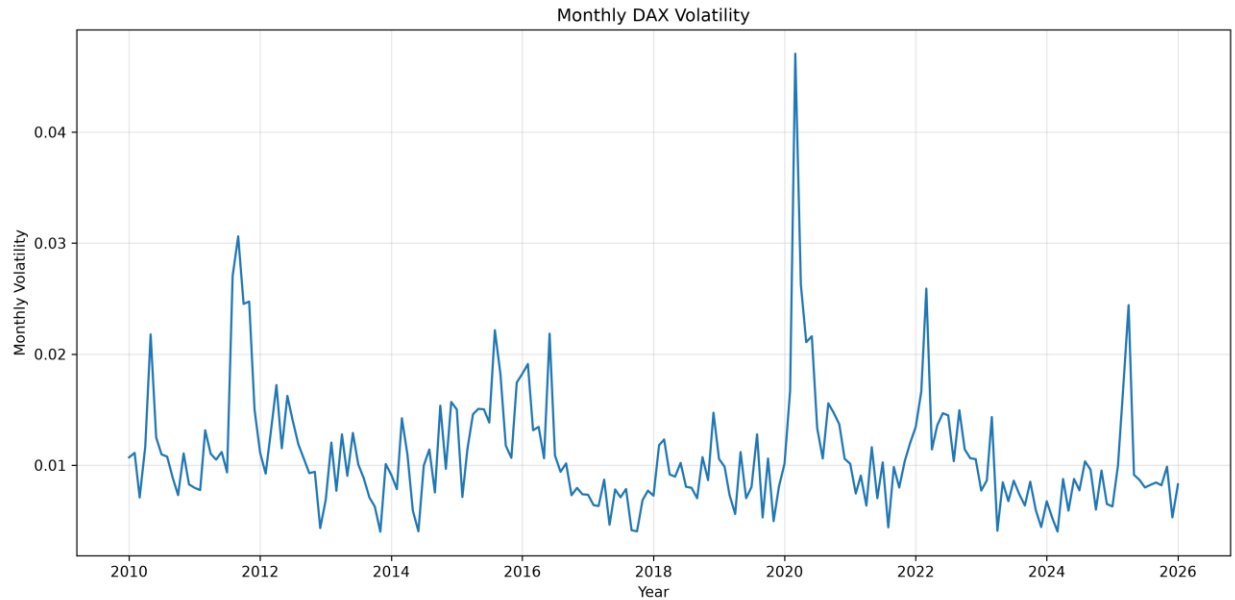


Figure 3 presents monthly DAX Volatility over the sample period. Volatility tends to increase during periods of financial and economic stress. The most pronounced spike occurred during the COVID-19 crisis in 2020, reflecting heightened uncertainty and instability in financial markets. Additional increases are visible during the Eurozone debt crisis and the Ukraine war period, although these are less pronounced than the COVID-19 shock.

Note: Vertical dashed lines indicate major economic and geopolitical events associated with increased market uncertainty.

5. Preliminary Findings

The preliminary analysis provides initial evidence of a relationship between economic uncertainty and information-seeking behavior. Descriptive statistics reveal substantial variation in the Germany EPU Index over time, while the correlation analysis indicates a moderate positive association between economic policy uncertainty and Google search activity ($r = 0.626$). The visual analysis further suggests that major economic and geopolitical events are associated with increases in both uncertainty and search intensity.

Overall, these findings are consistent with the hypothesis that individuals seek more information during periods of heightened uncertainty. While the correlation analysis does not establish causality, it provides preliminary support for the proposed relationship between uncertainty and information-seeking behavior. The next stage of the project will employ regression analysis to formally examine whether increases in economic policy uncertainty led to higher search activity and changes in financial market volatility.